

2023 Ph H2 Q5

Section: Particles and Waves

Topic: Doppler Effect & Semiconductors

Summary of Question:

A siren at 440 Hz moves at 31 m s^{-1} . Speed of sound = 340 m s^{-1} .

(a)(i) Doppler frequency approaching

$$f' = f \times v / (v - v_s) = 440 \times 340 / 309 = 484 \text{ Hz}$$

Answer: 484 Hz

(a)(ii) Frequency receding

Frequency is less than the source frequency.

Wavefronts further apart → lower frequency.

Answer: Less than

(b)(i) Period of AC supply

$$f = 2 \text{ Hz} \rightarrow T = 1/f = 0.5 \text{ s}$$

Answer: 0.5 s

(b)(ii) LEDs alternating

LEDs conduct in one direction only.

Red LEDs conduct on one half-cycle, blue on the other.

Answer: Opposite biasing, so not lit simultaneously

(b)(iii)(A) Photon energy

$$E = hc/\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(625 \times 10^{-9}) = 3.18 \times 10^{-19} \text{ J}$$

Answer: $3.18 \times 10^{-19} \text{ J}$

(b)(iii)(B) Band gap

Blue LEDs have larger band gap $\rightarrow$ photons have higher frequency, shorter wavelength, more energy.

Answer: Blue > Red photon energy

Final Answers

(a)(i) 484 Hz

(a)(ii) Less than

(b)(i) 0.5 s

(b)(ii) Opposite biasing, only one colour conducts at a time

(b)(iii)(A) $3.18 \times 10^{-19} \text{ J}$

(b)(iii)(B) Blue LEDs: larger band gap, higher photon energy

Revision Tips

- Doppler effect formula: $f' = f \times v/(v \mp v_s)$.

- Diodes conduct only when forward biased.
- Photon energy inversely proportional to wavelength.